A GENERICALLY FINITE MORPHISM OVER A FINITE GENERIC FIBRE IS FINITE OVER A DENSE OPEN OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 13328-13339 and proof span 13341-13367 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the two alternatives in the hypothesis and treats them in the separated and quasi-compact cases; the author's wording is preserved.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The generic finiteness statement producing a dense open of the target and an open of the source over which the restriction is finite, cited from the same chapter. Cited at proof lines 13345-13346. Provides the pair of opens in the separated case, which then has to be proved to exhaust the preimage.
- A finite morphism is closed and proper, and the image of a proper scheme over the target is closed. Cited at proof lines 13347-13350. Shows that the open piece is also closed inside the preimage of the chosen open, which forces the complementary open to be empty.
- A quasi-compact scheme contains a dense open separated subscheme, and for a quasi-compact morphism whose image of a closed subset misses a point there is an open neighbourhood of that point whose preimage misses that subset, both cited from the properties chapter and the same chapter. Cited at proof lines 13356-13364. Reduces the quasi-compact case to the separated case by first restricting to a separated dense open of the source.

Source reference. Original lemma 01WN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2488–2488. Same-chapter reference used in the retained author text at source line(s) 11282, 12507, 13352. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers

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the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 02NW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3495–3495. Same-chapter reference used in the retained author text at source line(s) 13285, 13296, 13349, 13488, 13520. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01W6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2471–2471. Same-chapter reference used in the retained author text at source line(s) 10104, 10627, 10794, 10905, 11283, 13351, 14146. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01RL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2306–2306. Same-chapter reference used in the retained author text at source line(s) 1323, 1327, 1371, 13364. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03J1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4623–4623. Same-chapter reference used in the retained author text at source line(s) 13360. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X, Y be schemes. Let $f : X \to Y$ be locally of finite type. Let X^0, resp. Y^0 denote the set of generic points of irreducible components of X, resp. Y. Assume*

(1) *X^0 and Y^0 are finite and $f^{-1}(Y^0) = X^0$,*
(2) *either f is quasi-compact or f is separated.*

Then there exists a dense open $V \subset Y$ such that $f^{-1}(V) \to V$ is finite.

Proof. Since Y has finitely many irreducible components, we can find a dense open which is a disjoint union of its irreducible components. Thus we may assume Y is irreducible affine with generic point η. Then the fibre over η is finite as X^0 is finite.

Assume f is separated and Y irreducible affine. Choose $V \subset Y$ and $U \subset X$ as in Lemma 02NW part (3). Since $f|_U : U \to V$ is finite, we see that $U \subset f^{-1}(V)$ is closed as well as open (Lemmas 01W6 and 01WN). Thus $f^{-1}(V) = U \amalg W$ for some open subscheme W of X. However, since U contains all the generic points of X we conclude that $W = \emptyset$ as desired.

Assume f is quasi-compact and Y irreducible affine. Then X is quasi-compact, hence there exists a dense open subscheme $U \subset X$ which is separated (Properties, Lemma 03J1). Since the set of generic points X^0 is finite, we see that $X^0 \subset U$. Thus $\eta \notin f(X \setminus U)$. Since $X \setminus U \to Y$ is quasi-compact, we conclude that there is a nonempty open $V \subset Y$ such that $f^{-1}(V) \subset U$, see Lemma 01RL. After replacing X by $f^{-1}(V)$ and Y by V we reduce to the separated case which we dealt with in the preceding paragraph. □